Module 3: Data Link Layer and Wireless Networks

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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

Virtual LANs (VLANs)

- Traditional LAN membership is based on physical location.
- VLAN (Virtual LAN) groups devices logically rather than physically.
- VLANs are created through software configuration.
- Physical rewiring is not required when users change groups.

Definition

A VLAN is a logical network created through software configuration rather than physical wiring.

Need for VLANs

In traditional switched LANs:

- Workgroups are based on physical network connections.
- Changing departments or projects requires rewiring.
- Reconfiguration increases cost and administration effort.

Problem

Physical movement of users often requires physical network changes.

Traditional LAN vs VLAN

Traditional LAN	VLAN
Physical grouping	Logical grouping
Requires rewiring	Software configuration
Less flexible	Highly flexible
Geographic membership	Functional membership

VLAN Concept

A physical LAN can be divided into multiple logical networks called VLANs.

Characteristics:

- Each VLAN represents a workgroup.
- Members communicate as if connected to the same LAN.
- Stations can be moved between VLANs using software.
- Broadcast traffic remains within the VLAN.

VLAN Broadcast Domains

- Each VLAN forms a separate broadcast domain.
- Broadcast messages are delivered only to members of the same VLAN.
- Stations outside the VLAN do not receive these broadcasts.

Key Idea

VLANs divide one physical LAN into multiple logical broadcast domains.

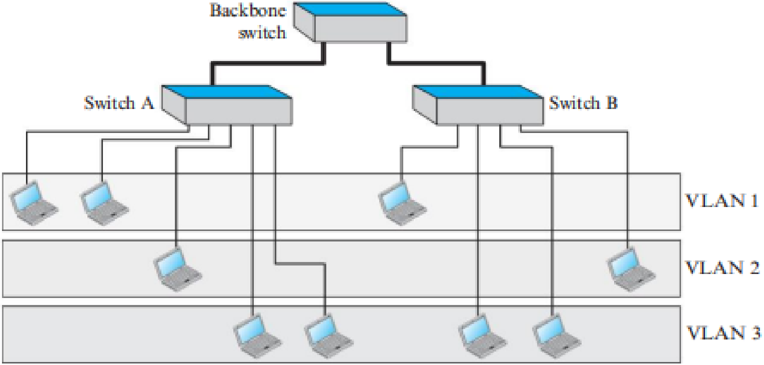
VLAN Across Multiple Switches

VLAN technology allows:

- Users connected to different switches
- Users in different buildings
- Users on separate physical LANs

to belong to the same VLAN.

Figure 5.60 *Two switches in a backbone using VLAN software*



VLAN Membership Criteria

Different vendors use different characteristics to define VLAN membership.

Common methods:

- Interface Numbers
- MAC Addresses
- IP Addresses
- Multicast IP Addresses
- Combination of Multiple Criteria

Membership Using Interface Numbers

Membership is determined by switch port numbers.

Example:

- Ports 1, 2, 3, 7 $\rightarrow$ VLAN 1
- Ports 4, 10, 12 $\rightarrow$ VLAN 2

Easy to configure and widely used.

Membership Using MAC Addresses

Membership is determined by Ethernet MAC addresses.

Example:

- E2:13:42:A1:23:34 $\rightarrow$ VLAN 1
- F2:A1:23:BC:D3:41 $\rightarrow$ VLAN 1

Advantage:

- Device remains in the same VLAN regardless of switch port.

Membership Using IP Addresses

Membership is determined by network-layer addresses.

Example:

- 181.34.23.67
- 181.34.23.72
- 181.34.23.98

belong to the same VLAN.

Useful for network-based grouping.

Membership Using Multicast IP Addresses

Membership is based on multicast groups.

Characteristics:

- IP multicast groups are mapped to VLANs.
- Users receiving the same multicast traffic can be grouped together.
- Efficient for collaborative applications.

Combination-Based Membership

Modern VLAN software allows multiple criteria.

Examples:

- MAC + IP Address
- Port Number + MAC Address
- Department + Project Number

Provides greater flexibility and control.

VLAN Configuration Methods

Stations can be assigned to VLANs using:

1. Manual Configuration
2. Automatic Configuration
3. Semiautomatic Configuration

Manual Configuration

Characteristics:

- Administrator assigns devices manually.
- VLAN software is used.
- Membership changes require administrator intervention.

Advantages:

- Simple
- High administrative control

Automatic Configuration

Characteristics:

- VLAN assignment occurs automatically.
- Based on predefined rules.
- Users migrate automatically between VLANs.

Example:

- Project assignment determines VLAN membership.

Semiautomatic Configuration

A combination of manual and automatic approaches.

Typical operation:

- Initial setup performed manually.
- Future migrations handled automatically.

Provides a balance between control and flexibility.

Communication Between Switches

In a multi-switch VLAN environment:

- Each switch must know VLAN membership information.
- Membership data must be exchanged between switches.

Methods used:

1. Table Maintenance
2. Frame Tagging
3. Time Division Multiplexing (TDM)

Table Maintenance Method

- Switch records VLAN membership information.
- Entries are stored in a table.
- Switches periodically exchange tables.
- Membership information remains synchronized.

Frame Tagging Method

- Extra VLAN header added to frames.
- VLAN identifier travels with the frame.
- Receiving switch reads VLAN tag.
- Frame delivered to correct VLAN.

Most Popular Method

Frame tagging is the basis of IEEE 802.1Q.

Time Division Multiplexing (TDM)

- Trunk links divided into multiple channels.
- Each channel assigned to a VLAN.
- Frames travel through VLAN-specific channels.

Example:

- Channel 1 $\rightarrow$ VLAN 1
- Channel 2 $\rightarrow$ VLAN 2
- Channel 3 $\rightarrow$ VLAN 3

IEEE 802.1Q Standard

In 1996, IEEE introduced:

$\boxed{\text{IEEE 802.1Q}}$

Functions:

- Standardized VLAN frame tagging.
- Supports multivendor equipment.
- Enables VLAN interoperability.
- Defines trunk communication standards.

Advantages of VLANs

Major benefits include:

- Reduced network management cost.
- Easier user migration.
- Flexible workgroup creation.
- Better traffic management.
- Enhanced scalability.
- Improved security.

Cost and Time Reduction

Without VLANs:

- Physical rewiring required.
- Increased maintenance effort.
- Higher operational cost.

With VLANs:

- Software-based changes.
- Faster configuration.
- Lower cost.

Creating Virtual Work Groups

VLANs enable users with common tasks to work together.

Examples:

- Research teams
- Project groups
- Faculty committees
- Development teams

Physical location becomes irrelevant.

Security Benefits of VLANs

VLANs improve security by isolating traffic.

Benefits:

- Broadcast messages remain inside VLAN.
- Unauthorized users cannot receive VLAN traffic.
- Better separation of departments and projects.
- Reduced risk of information leakage.

Summary of VLANs

- VLANs create logical networks using software.
- Membership can be based on ports, MAC addresses, or IP addresses.
- VLANs define separate broadcast domains.
- Devices can be moved between VLANs without rewiring.
- IEEE 802.1Q standardizes VLAN tagging.
- VLANs improve flexibility, security, and manageability.

Key Point

A VLAN groups users logically rather than physically, making modern networks more flexible and easier to manage.